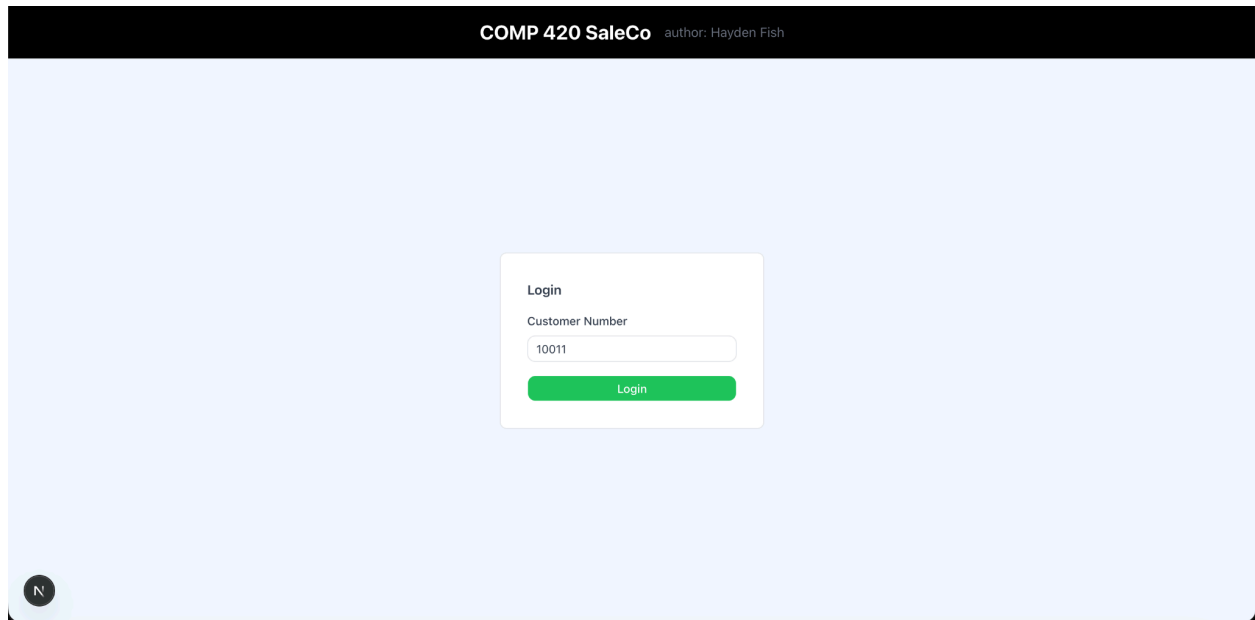


Part I



Part II

I designed a simple single-page web application using [Next.js](#) and React to simulate typical CRUD operations. The UI uses prebuilt components from the ShadCN library combined with TailwindCSS.

The database logic happens on a python backend, running on a server using FastAPI (uvicorn). MySQL Connector is utilized for database queries.

For directions to **run the application** see the [README.md](#) in the root of the project folder.

GET

COMP 420 SaleCo

author: Hayden Fish

User Guide

Customers

Customer

First Name	Last Name	Initial	Area Code	Number	Balance	Actions
Alfred	Ramas	A	615	844-2573	0	EditDelete
Leona	Dunne	K	713	894-1238	615.73	EditDelete
Kathy	Smith	W	615	894-2285	0	EditDelete
Paul	Olowski	F	615	894-2180	0	EditDelete
Myron	Orlando		615	222-1672	0	EditDelete
Amy	O'Brian	B	713	442-3381	0	EditDelete
James	Brown	G	615	297-1228	277.55	EditDelete
George	Williams		615	290-2556	0	EditDelete
Anne	Farriss	G	713	382-7185	0	EditDelete
Olette	Smith	K	615	297-3809	0	EditDelete

N

Transactions

Transaction

POST

COMP 420 SaleCo

author: Hayden Fish

User Guide

Customers

Customer

First Name	Last Name	Initial	Area Code	Number	Balance	Actions
Alfred	Ramas	A	615	844-2573	0	EditDelete
Leona	Dunne	K	713	894-1238	615.73	EditDelete
Kathy	Smith	W	615	894-2285	0	EditDelete
Paul	Olowski	F	615	894-2180	0	EditDelete
Myron	Orlando		615	222-1672	0	EditDelete
Amy	O'Brian	B	713	442-3381	0	EditDelete
James	Brown	G	615	297-1228	277.55	EditDelete
George	Williams		615	290-2556	0	EditDelete
Anne	Farriss	G	713	382-7185	0	EditDelete
Olette	Smith	K	615	297-3809	0	EditDelete

N

Transactions

Transaction

Create Customer (by: Hayden Fish)

Number

10050

First Name

John

Last Name

Doe

Initial

J

Area Code

805

Phone

123-4567

Balance

150.87

Password

.....

Create

James	Brown	G	615	297-1228	277.55	EditDelete
George	Williams		615	290-2556	0	EditDelete
Anne	Farriss	G	713	382-7185	0	EditDelete
John	Doe	J	805	123-4567	151	EditDelete

PUT

COMP 420 SaleCo

author: Hayden Fish

User Guide

Customers

Customer

First Name	Last Name	Initial	Area Code	Number	Balance	Actions
Ian	Fish	H	805	5393231	150000	EditDelete
Hayden	Fish	H	123	456-7890	150	EditDelete
Leona	Dunne	K			615.73	EditDelete
Kathy	Smith	W				EditDelete
Paul	Olowski	F				EditDelete
Myron	Orlando					EditDelete
Amy	O'Brian	B				EditDelete
James	Brown	G			277.55	EditDelete
George	Williams					EditDelete
Anne	Farriss	G				EditDelete
John	Doe	J			51	EditDelete

Transaction

Transaction #	Date	Amount	Actions
10003	2023-01-17T00:00:00	329.66	EditDelete
10004	2023-01-17T00:00:00	615.73	EditDelete
10006	2023-01-29T00:00:00	329.66	EditDelete
10007	2023-01-18T00:00:00	277.55	EditDelete
10010	2026-04-28T00:00:00	500	EditDelete

Invoices

N

Invoice #	Customer #	Date	Subtotal	Tax	Total	Payment Type	Payment Amount	Balance	Actions
1001	10014	2023-01-16T00:00:00	54.92	4.39	59.31	cc	59.31	0	EditDelete

Edit Customer

First Name

John

Last Name

Doe

Initial

J

Area Code

805

Phone

123-4567

Balance

151

Password

.....

Save Changes

DELETE

Anne	Farriss	G	713	382-7185	0	EditDelete
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[User Guide](#)

Customers

[Customer](#)

First Name	Last Name	Initial	Area Code	Number	Balance	Actions
Leona	Dunne	K	713	894-1238	615.73	Edit Delete
Kathy	Smith	W	615	894-2285	0	Edit Delete
Paul	Olowski	F	615	894-2180	0	Edit Delete
Myron	Orlando		615	222-1672	0	Edit Delete
Amy	O'Brian	B	713	442-3381	0	Edit Delete
James	Brown	G	615	297-1228	277.55	Edit Delete
George	Williams		615	290-2556	0	Edit Delete

Selected item is removed from the database

Part III

RESTful API handling database transactions

Smith	W	615		894-2285	0	Edit Delete
Olowski	F	615		894-2180	0	Edit Delete
Orlando		615		222-1672	0	Edit Delete
O'Brian	B					Edit Delete
Brown	G				5	Edit Delete
Williams						Edit Delete
Date				Amount		Actions
2023-01-17T00:00:00				29.66		Edit Delete
2023-01-17T00:00:00				15.73		Edit Delete
2023-01-29T00:00:00				29.66		Edit Delete
2023-01-18T00:00:00				77.55		Edit Delete
Date	Subtotal	Tax	Total	Payment Type	Payment Amount	Balance
2023-01-16T00:00:00	54.92	4.39	59.31	cc	59.31	0

User Guide (by: Hayden Fish)

Part I

Login to the application with your **customer code** as per instructions.

Part II

Each section of the page shows a table of records - Customers, Transactions, Invoices, Lines, Products, and Vendors. Scroll down to browse all sections. **CRUD** operations are available for each table and their respective models. Appropriate error messages are featured for each request.

Part III

HTTP requests are made to a backend which queries the database and persists all transactions. In the case of constraints, such as foreign keys when deleting records, all dependent records will also be removed.

OpenAPI specs

FastAPI 0.1.0 OAS 3.1
/openapi.json

default ^		
GET	/api/vendors/all	Get All Vendors
POST	/api/vendors/save	Save Vendor
PUT	/api/vendors/update/{id}	Update Vendor
DELETE	/api/vendors/delete/{vendor_id}	Delete Vendor By Id
GET	/api/products/all	Get All Products
POST	/api/products/save	Save Product
PUT	/api/products/update/{id}	Update Product
DELETE	/api/products/delete/{product_id}	Delete Product By Id
GET	/api/lines/all	Get All Lines
POST	/api/lines/save	Save Line
PUT	/api/lines/update/{invoice_id}/{line_id}	Update Line
DELETE	/api/lines/delete/{invoice_id}/{line_id}	Delete Line By Id
GET	/api/invoices/all	Get All Invoices
POST	/api/invoices/save	Save Invoice
PUT	/api/invoices/update/{id}	Update Invoice
DELETE	/api/invoices/delete/{invoice_id}	Delete Invoice By Id
GET	/api/transactions/all	Get All Transactions
POST	/api/transactions/save	Save Transaction
PUT	/api/transactions/update/{id}	Update Transaction